www.csvtuonline.com

Roll No. .....

324453(25)

B. E. (Fourth Semester)

EXAMINATION, April-May, 2015

(New Scheme)

(Branch : Elect.)

NETWORK ANALYSIS AND SYNTHESIS

Time : Three Hours] [*Maximum Marks : 80*

[*Minimum Pass Marks : 28*

Note : Part (a) of each question is compulsory. Solve any *two* from Part (b), (c) and (d).

Unit—I

1. (a) Define impulse function. 2

(b) Find the Laplace transform of the waveform shown in Fig. 7

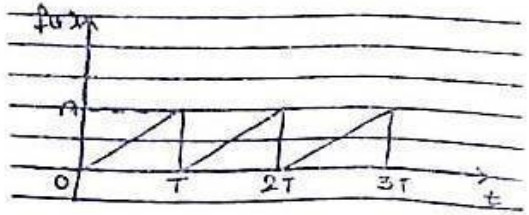


Fig. 1

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(c) For the network shown in Fig., find expression for $v_0(t)$.

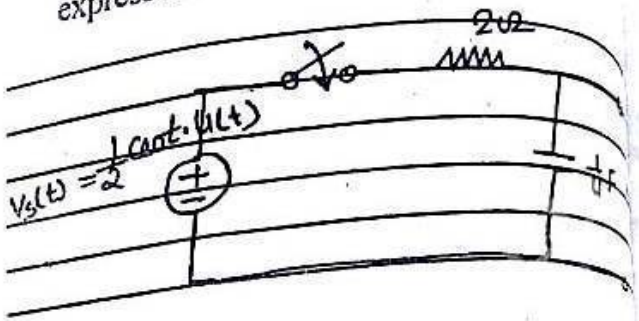


Fig. 2

(d) In the network of Fig. shown below, the switch changed from the position a to b at $t = 0$. Solve for $i, di/dt$ and $\frac{d^2i}{dt^2}$ at $t = 0^+$.

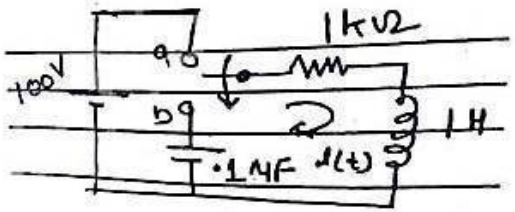


Fig. 3

Unit—II

2. (a) Define poles and zeros. 2

(b) The network in the figure contains a resistor and a capacitor in addition to various sources. With respect to the load consisting of RL in series

www.csvtuonline.com

www.csvtuonline.com

with L, determine the Thevenin equivalent network. 7

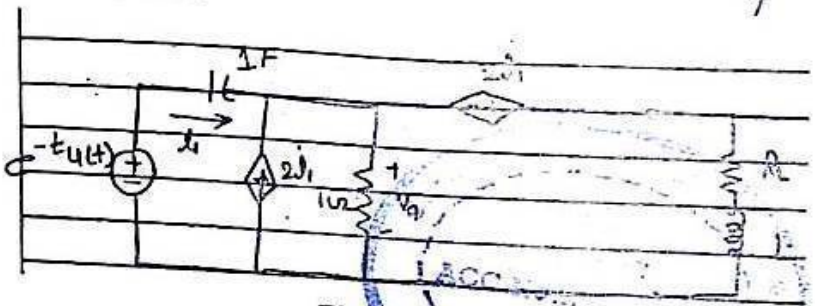


Fig. 4

(c) For the resistive two-port network of the figure, determine the numerical value of (i) G_{12} (ii) Z_{12} (iii) Y_{12} (iv) α_{12}. 7

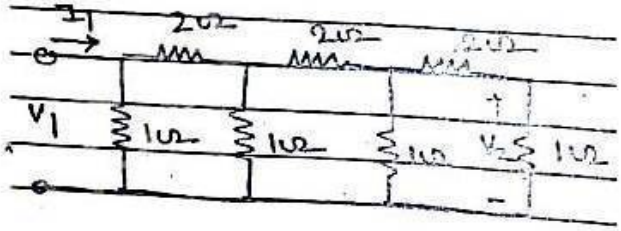


Fig. 5

(d) Find $I_1(t)$ resulting from closing the switch at $t = 0$ with the circuit previously unenergized. The circuit constants are $L_1 = 1$ H, $L_2 = 4$ H, $M = 2$ H, $R_1 = R_2 = 1\Omega$, $V = 1$ V. 7

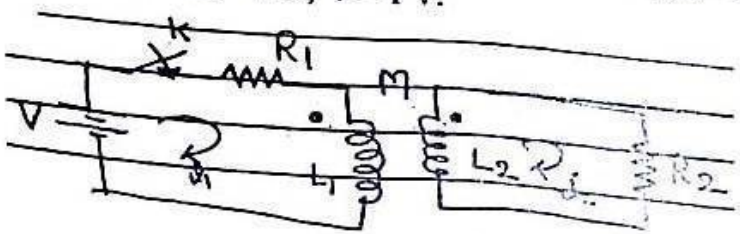


Fig. 6

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

Unit—III

3. (a) Explain the condition of reciprocity for A parameter.

(b) Find Z-parameter for the network shown in below. Find h-parameters.

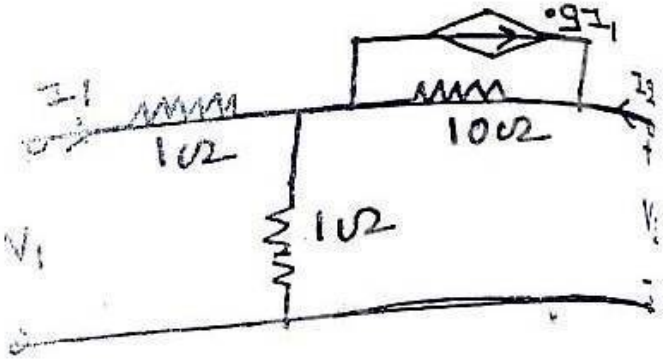


Fig. 7

(c) The ABCD parameters of a two-port network shown in Fig. below have A = 2.5, B = 40, C = 1℧, D = 2. What must be the input voltage V_1 applied for the output voltage V_2 to be 10 across the load of 10 Ω connected at port 2.

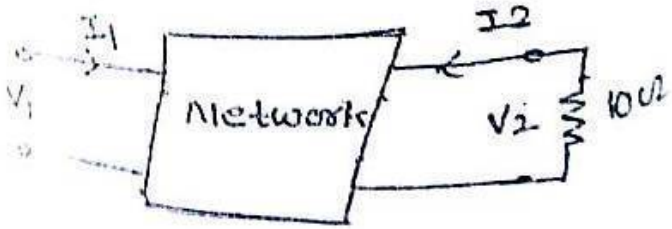


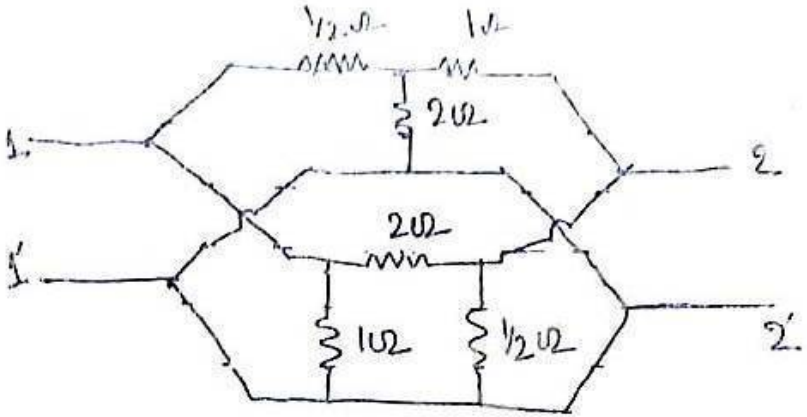
Fig. 8

www.csvtuonline.com

www.csvtuonline.com

[5]

(d) [illegible] 7



Unit—IV

4. (a) Define Hurwitz Polynomials. 2

(b) Test whether the polynomial is Hurwitz : 7

(i) $P_{(s)} = s^3 + 4s^2 + 5s + 2$

(ii) $P_{(s)} = s^4 + s^3 + 3s^2 + 2s + 12$

(c) Realise the foster 1st form and Cauer 2nd form of the impedance function : 7

$$z(s) = \frac{(s+1)(s+3)}{s(s+2)}$$

(d) (i) Explain "Positive Real function" and its properties. 3

(ii) Test whether $F(s) = \frac{s^3 + 6s^2 + 7s + 3}{s^2 + 2s + 1}$ is positive real function. 4

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

Unit—V

5. (a) Explain filter classification.

(b) A π-section filter network consists of a [illegible] arm inductor of 20 mH shunt-arm capacit[illegible] .16 μF each. Calculate the cut-off frequency, [illegible] attenuation and phase shift at 15 kHz. What is [illegible] value oft he nomial impedance in the pass ba[illegible]

(c) Design an *m*-derived T-section low-pass [illegible] having cut-off frequency of 800 Hz, de[illegible] impedance of 500 Ω and frequency [illegible] attenuation of 1000 Hz.

(d) Design the π-section of an *m*-derived high-[illegible] filter having a design impedance of 60[illegible] cut-off frequency of 4 kHz and an inf[illegible] attenuation at 3.6 kHz.

www.csvtuonline.com